Practical 17 on Generalised Linear Models (GLMs)

Parkrun (https://www.parkrun.org.uk/) is a non-competitive event held across the UK (and now globally) every saturday where people are invited to run an approximate 5km route. There are now several hundred locations where these events are held.

The organisers of each event would like to have a good idea ahead of the event of the number of runners to assist with logistics. The organisers are all volunteers, and on a busy day require more people to help with 'barcode collection' which is used to calculate people's runtimes.

Data are avaialable from a sample of parkrun events held between 2014-2016. The data include the number of people who attended, grouped by age and sex. Additionally, data are available on meterological (weather) variables, as it has been hypothesised that the weather influences the number of runners in attendance. The varaibles of interest are as follows;

- **total** represent the total number of runners that attend each event
- **sm20-24** to **vw70_74** represent the number of runners in each category 'senior men (sm)', 'senior women (sw)', 'vetran men (vm)', 'vetran women (vm)', and the more specifc age category (in 5 years age bands)
- **loc** location of the event (4 locations)
- **date** date of the event
- **rh** relative humidity, which is a reliable indicator of whether it is raining. Higher values of rh indicate that it is raining
- **rhc** is relative humidity, centered on 70 (more than this meas rain is likely)
- **temp** temperature (degrees celcius)
- **tempc** is temperature, centered on 10oC
- **wind speed** and **wind direction** are included as variables, but are largely missing, so should be excluded from the analysis

Part 1: Data exploration

Task: Use appropriate graphical tools and summaries to explore the variables in the data and produce a table of baseline characteristics of parkrun attendance and other variables.

Hint: you should also consider the timing of the parkrun events. For example, what impact does the month of the year have on attendance?

Try not to spend too long on this in the session, we only have 90mins!

In [18]:
```
# load relevant libraries
library(ggplot2)
install.packages("sandwich")
library(sandwich)

setwd("~/Dropbox/Dropbox/Teaching-LSHTM/MSc - DataScience/SHDS-module/
#dir()
# load in the data - returns the dataset 'dat'
load("parkrun_attend_clean_2015.Rdat")
dim(dat)
```

```
The downloaded binary packages are in
        /var/folders/jp/f50wn8ds35b_c2ns4tld6v6h0000gr/T//Rtmps67b5
W/downloaded_packages

539  39
```

In [19]:
```
head(dat)
table(dat$loc)
```

A data.frame: 6 × 39

	loc	date	rh	wet_bulb	temp	wind_speed	wind_direction	sm20_24	sr
	<chr>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	
41	brockwell	01/04/2017	72.4	0	12.9	NA	NA	2	
43	crystalpalace	01/04/2017	81.3	0	11.0	12	210	1	
44	finsbury	01/04/2017	72.4	0	12.9	NA	NA	0	
48	kingston	01/04/2017	76.5	0	12.4	NA	NA	0	
67	brockwell	01/07/2017	72.2	0	16.1	NA	NA	2	
69	crystalpalace	01/07/2017	77.1	0	16.0	9	300	1	

```
    brockwell crystalpalace      finsbury      kingston
          130           136           128           145
```

In [20]:
```
print("These data should have not missing data")
summary(dat$tempc)  #
summary(dat$rhc)  #
```

```
[1] "These data should have not missing data"

   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-12.000  -2.350   2.000   1.673   6.000  13.700

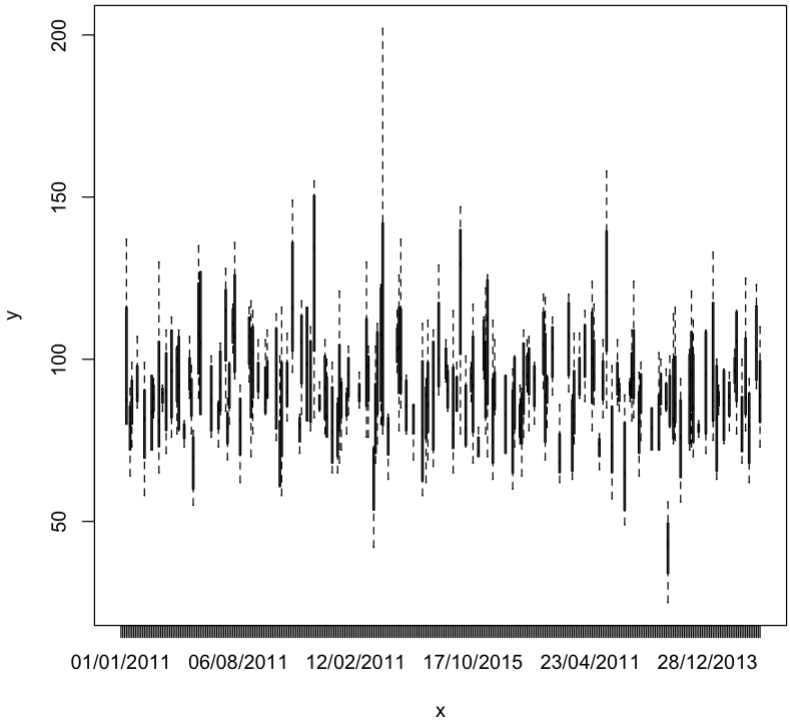
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-21.100  -0.850   8.300   8.197  17.800  30.000
```

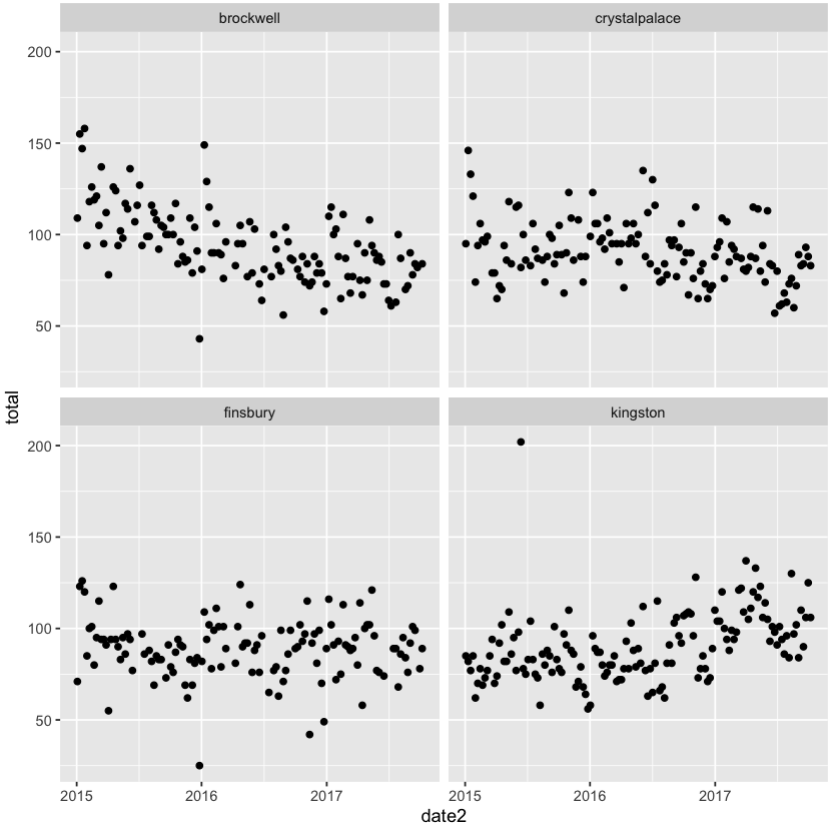
In [22]:
```
# plot the data - total number that attend

par(mfrow=c(1,1))
plot(dat$date,dat$total) # can't see much going on here...

# but, location might be important?
ggplot(dat,aes(x=date2,y=total,group=loc)) + geom_point() + facet_wrap

# decide on suitable summary stats...
```





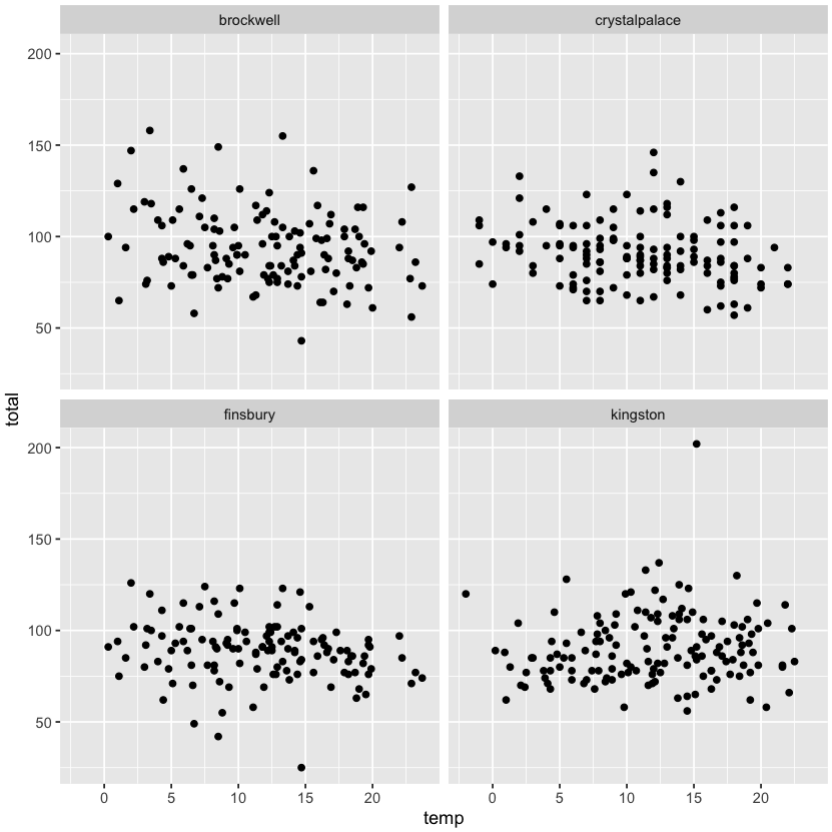
Part 2: Univariable analysis.

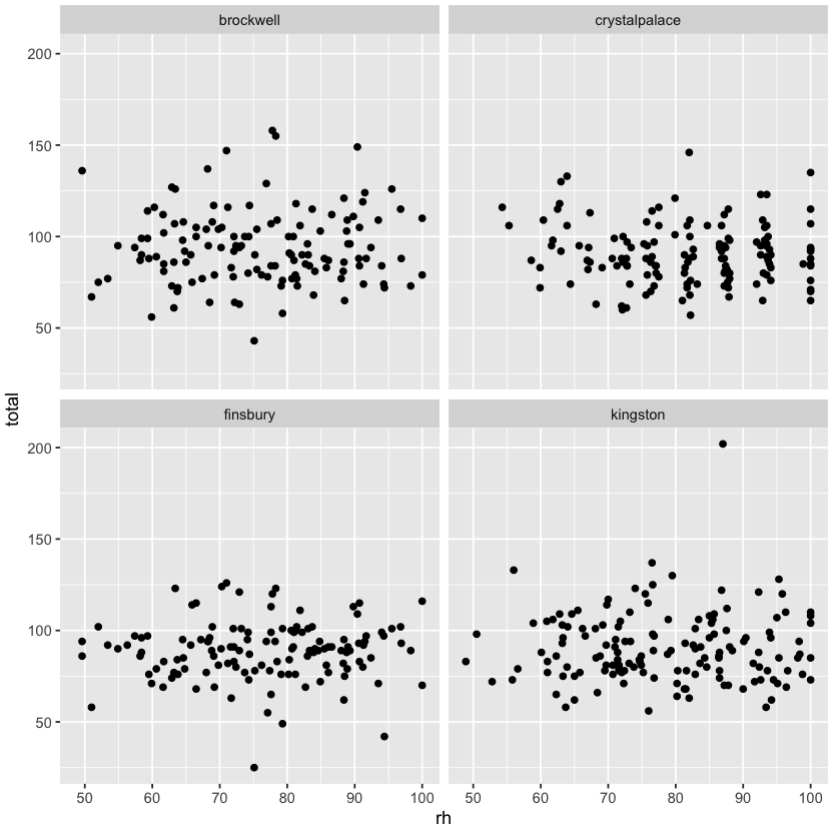
Task: Use a Poisson GLM to estimate the effect of the weather and other variables on parkrun attendance.

In [23]:
```
# plot first...
# what is the relationship between temp and rh on total attendance. Is

ggplot(dat,aes(x=temp,y=total)) + geom_point() + facet_wrap(~loc)

ggplot(dat,aes(x=rh,y=total)) + geom_point() + facet_wrap(~loc)
```





Notes:

- A linear trend for temp & rh is not very strong if it is even present. Scatter plots indicate only a slight trend.
- Effect of rh on total is more obvious. It might be worth considering interactions between location and rh.

In [24]:
```
# part 2: GLMs - univariable

m1 <- glm(total ~ 1,family="poisson", data=dat)
summary(m1)

# this would be the 'default' model, and the effect of rh examined in
# Residual deviance:
# The residual deviance is the difference between the deviance of the
# the maximum deviance of the ideal model where the predicted values a
# the observed. Therefore, if the residual difference is small enough,
# fit test will not be significant, indicating that the model fits the

m1a <- glm(total ~ year,family="poisson", data=dat)
summary(m1a)

m1b <- glm(total ~ loc,family="poisson", data=dat)
summary(m1b)

m1c <- glm(total ~ tempc,family="poisson", data=dat)
summary(m1c)
print("Point estimates")
(exp(coefficients(m1c)))
exp(confint(m1c))

m1d <- glm(total ~ rhc,family="poisson", data=dat)
summary(m1d)
```

```
Call:
glm(formula = total ~ 1, family = "poisson", data = dat)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) 4.510860   0.004515     999   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 1925.6  on 538  degrees of freedom
Residual deviance: 1925.6  on 538  degrees of freedom
AIC: 5339.9

Number of Fisher Scoring iterations: 4


Call:
```

Notes:

> Univariable analysis:

Model	deviance	df.	AIC	p-value of Wald statistic
Null.	1925.6	538	5339.9	n/a
Year.	1904.5	536	5322.8	<0.001
Loc.	1904.0	535	5324.3	<0.01
Temp.	1897.7	537	5313.9	<0.01
Rh.	1925.6	537	5341.9	0.963

> In the regression model it is a 'Wald statistic' that is being applied to each explanatory variable to test for its statistical significance. (Note that any more information about the Wald statistic is not necessary for interpretation)

> Each variable we've trailled in the model, with the exception of Rh, the 95% CI of the coefficients having consistently positive or negative value, and the p-values suggest evidence for each variable being associated with total attendance.

Part 3: Checking assumptions of your model.

Task: Examine the models to see whether the data are Poisson distributed, or if an alternative model structure should be explored. Comment of the tests you used to make your decision.

Hint: some useful code are below for using robust standard errors. Note however, that either this option should be selected, **or** a quasi-poisson model specified, but not both!

In [27]:
```
pchisq(1925.6, 538, lower.tail=FALSE)
```

4.24505051356977e-155

The null hypothesis is that our model is correctly specified. The p-value returned suggests considerable evidence against the null.

Typically, overdispersion is present in the data. We can look for this using,

In [28]:
```
# what is the mean and variance?
print(mean(dat$total))
print(var(dat$total))
```

```
[1] 91
[1] 332.3086
```

As suspected, we have considerable over-dispersion. One option to overcome this issue is to use 'robust standard errors'. see below,

In [30]:
```
#library(sandwich)

cov <- vcovHC(m1, type="HC0")
se <- sqrt(diag(cov))
robust.SE <- cbind(Estimate= coef(m1), "Robust SE" = se,
                   "Pr(>|z|)" = 2 * pnorm(abs(coef(m1)/se),lower.tail=
                   LL = coef(m1) - 1.96 * se,
                   UL = coef(m1) + 1.96 * se)
print(robust.SE)
summary(m1)
# for both these models we can see the residual deviance is quite a bi
# than the null deviance, indicating the model isn't great.
# additionally, the residual deviance is larger than the degrees of fr
```

```
            Estimate   Robust SE Pr(>|z|)       LL       UL
(Intercept)  4.51086 0.008620486        0 4.493963 4.527756


Call:
glm(formula = total ~ 1, family = "poisson", data = dat)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) 4.510860   0.004515     999   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 1925.6  on 538  degrees of freedom
Residual deviance: 1925.6  on 538  degrees of freedom
AIC: 5339.9

Number of Fisher Scoring iterations: 4
```

> Another option is to consider a model that better accounts for overdispersion, such as a quasipoisson model.

In [33]:
```
# examine a quasi-poisson model
m2 <- glm(total ~ 1,family="quasipoisson", data=dat)
summary(m2)
```

```
Call:
glm(formula = total ~ 1, family = "quasipoisson", data = dat)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 4.510860   0.008628   522.8   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for quasipoisson family taken to be 3.651742)

    Null deviance: 1925.6  on 538  degrees of freedom
Residual deviance: 1925.6  on 538  degrees of freedom
AIC: NA

Number of Fisher Scoring iterations: 4
```

Notes:

> The goodness of fit test indicates evidence of a poorly fitting model. Calculation of robust stnd errors alters the 95% CI of the coefficients.

> Examining the quasipoisson model illustrates that the over-dispersion parameter is required in the model. The estimate of overdispersion in ~4. The summary statistics of the data show that the variance is quite a larger than the mean, so a quasipoisson model would be more appropriate.

Part 4: Multivariable analysis.

Task: Construct a multivarable model of *location, month, temp, and humidity* that could be used to predict parkrun attendance.

In [34]:
```
m2a <- glm(total ~ rhc + tempc + year + loc + month,family="quasipoiss
summary(m2a)  # no impact of rh

# output robust std errors
cov <- vcovHC(m2a, type="HC0")
se <- sqrt(diag(cov))
robust.SE <- cbind(Estimate= coef(m2a), "Robust SE" = se,
                   "Pr(>|z|)" = 2 * pnorm(abs(coef(m2a)/se),lower.tail
                   LL = round(coef(m2a) - 1.96 * se,3),
                   UL = round(coef(m2a) + 1.96 * se,3))
print(robust.SE)

# check assumptions
pchisq(m2a$deviance, m2a$df.residual, lower.tail=FALSE)  # p<0.01 idea

# so assumptions are a bit violated
# use package 'sandwich' to adjust the standard errors to 'robust' sta
```

```
Call:
glm(formula = total ~ rhc + tempc + year + loc + month, family = "qu
asipoisson",
    data = dat)

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)       4.6838427  0.0332074 141.048  < 2e-16 ***
rhc              -0.0004716  0.0009396  -0.502  0.61597
tempc            -0.0019581  0.0028186  -0.695  0.48754
year2016         -0.0405638  0.0195483  -2.075  0.03847 *
year2017         -0.0334070  0.0210847  -1.584  0.11371
loccrystalpalace -0.0367790  0.0236959  -1.552  0.12124
locfinsbury      -0.0594888  0.0233012  -2.553  0.01096 *
lockingston      -0.0361426  0.0226449  -1.596  0.11108
month02          -0.0877310  0.0356644  -2.460  0.01422 *
month03          -0.1092387  0.0387827  -2.817  0.00504 **
month04          -0.0918128  0.0387081  -2.372  0.01806 *
month05          -0.0579681  0.0453201  -1.279  0.20144
month06          -0.0784568  0.0509016  -1.541  0.12384
month07          -0.1688860  0.0543704  -3.106  0.00200 **
month08          -0.1786112  0.0536348  -3.330  0.00093 ***
month09          -0.0877600  0.0451232  -1.945  0.05233 .
month10          -0.0974511  0.0419540  -2.323  0.02057 *
month11          -0.1756292  0.0426857  -4.114 4.51e-05 ***
month12          -0.2985858  0.0439660  -6.791 3.05e-11 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for quasipoisson family taken to be 3.190915)

    Null deviance: 1925.6  on 538  degrees of freedom
Residual deviance: 1632.8  on 520  degrees of freedom
AIC: NA

Number of Fisher Scoring iterations: 4
```

```
                     Estimate   Robust SE     Pr(>|z|)     LL     U
L
(Intercept)       4.6838426700 0.042529090 0.000000e+00  4.600  4.76
7
rhc              -0.0004715539 0.001053404 6.544073e-01 -0.003  0.00
2
tempc            -0.0019581413 0.002918979 5.023283e-01 -0.008  0.00
4
year2016         -0.0405638392 0.020101726 4.359874e-02 -0.080 -0.00
1
year2017         -0.0334070125 0.021518331 1.205448e-01 -0.076  0.00
9
loccrystalpalace -0.0367790420 0.021907253 9.318083e-02 -0.080  0.00
6
locfinsbury      -0.0594888198 0.021758876 6.256959e-03 -0.102 -0.01
7
lockingston      -0.0361425989 0.023478543 1.237095e-01 -0.082  0.01
0
month02          -0.0877309801 0.035424745 1.326608e-02 -0.157 -0.01
8
month03          -0.1092386753 0.039584154 5.786213e-03 -0.187 -0.03
2
month04          -0.0918128120 0.044761433 4.025147e-02 -0.180 -0.00
4
month05          -0.0579680954 0.049498756 2.415578e-01 -0.155  0.03
9
month06          -0.0784567811 0.070018687 2.624956e-01 -0.216  0.05
9
month07          -0.1688859866 0.061574720 6.092126e-03 -0.290 -0.04
8
month08          -0.1786111677 0.059193333 2.549360e-03 -0.295 -0.06
3
month09          -0.0877599636 0.046250385 5.776238e-02 -0.178  0.00
3
month10          -0.0974510517 0.043258703 2.427490e-02 -0.182 -0.01
3
month11          -0.1756291615 0.048081175 2.594332e-04 -0.270 -0.08
1
month12          -0.2985858489 0.048516646 7.541127e-10 -0.394 -0.20
3
```

4.21261706277471e-115

Notes:

> The model above combines several variables together and examines the effect of each variable independently. The variables year, month and location are all associated with the total attendance, as indicated by the 95% CIs. Referring to the robust stnd errors will be more appropriate than those directly from the model output.

> The model fit is still quite poor. So while we have improved the model assumptions by including a quasipoisson model sructure, it is likely that the assumptions of the model have been violated. Consequently, conclusions drawn from the model output should be treated with caution.

> Neither Rh or Temp are associated with a change in parkrun attendance. However, month is likely to be correlated with both these values. In terms of a hypothesis dirven approach, changes in monthly attendence may well be influenced by the weather, but in this current analysis we can't disentangle these effects.

Part 5: Model predictions.

Task: The parkrun organisers have requested that you predict the expected numbers of runners when the weather has the following conditions;

For each of the locations (Brockwell, Crystal Palace, Finsbury, Kingston):

- date: 01/05/2017
- temp: 10oC
- rh: 70

Comment on your findings. In the practical we will compare findings across each group, are some age-sex groups more strongly influenced by the weather?

Hint: Using the function predict() will return the expectation of the number of runners. For a prediction interval, we suggest sampling around this value to generate the distribution of runners.

In [36]:
```
# predict
# here we are using "new data" to create this prediction


ndat <- data.frame(expand.grid(loc=c("brockwell","crystalpalace","fins
                    tempc=c(10-10),rhc=c(70-70,55-70),
                    year=as.factor(2017),month=as.factor("05")))
#head(ndat)
tmp <- predict(m2a,newdata=ndat,se.fit=T)
ndat$pred <- round(exp(tmp$fit),3)
ndat$lwr <- round(exp(tmp$fit - tmp$se.fit*1.96),3)
ndat$upr <- round(exp(tmp$fit + tmp$se.fit*1.96),3)
#head(ndat)

# should also provide prediction intervals...
# for prediction intervals, we could simulate from the values assuming
ndat$pred.avg <- ndat$pred.lwr <- ndat$pred.upr <- 0
for(i in 1:dim(ndat)[1]){
  sim <- rpois(1000,rnorm(1000,ndat$pred[i],tmp$se.fit[i]))
  ndat$pred.avg[i] <- round(quantile(sim,0.5),2)
  ndat$pred.lwr[i] <- round(quantile(sim,0.025),2)
  ndat$pred.upr[i] <- round(quantile(sim,0.975),2)
}

print(ndat)
```

```
            loc tempc rhc year month   pred    lwr     upr pred.upr
pred.lwr
1     brockwell     0   0 2017    05 98.738 92.154 105.792   118.00
80.00
2 crystalpalace     0   0 2017    05 95.172 89.018 101.753   113.00
76.00
3      finsbury     0   0 2017    05 93.035 86.825  99.689   113.00
74.00
4      kingston     0   0 2017    05 95.233 88.990 101.913   115.00
74.98
5     brockwell     0 -15 2017    05 99.439 92.459 106.945   120.00
80.00
6 crystalpalace     0 -15 2017    05 95.848 88.963 103.266   116.02
77.00
7      finsbury     0 -15 2017    05 93.696 87.095 100.797   113.00
75.00
8      kingston     0 -15 2017    05 95.909 89.103 103.235   115.00
78.00
  pred.avg
1       98
2       94
3       93
4       95
5      100
6       96
7       93
8       96
```

Notes:

> The table provides estimates of the predictions for each location under the requested values. With the prediction intervals we are combining the uncertainty in the estimates of coefficients along with the variability of the underlying Poisson process. This means that prediction intervals are wider in value than model predictions.

> In this example we have retained the temp and rh values in the model predictions, but they really don't add much and could have been removed.